Simultaneous Equations and Inequalities

Section 1: Linear-Quadratic Simultaneous Systems

1 Solve the simultaneous system $y - 2x = 3$ and $x^2 + y^2 = 26$ for the positive solution pair (x, y).

a. $(1, 5)$

b. $(2, 7)$

c. $(3, 9)$

d. $(4, 11)$

2 Find the two points of intersection of the line $2x + y = 7$ and the rectangular hyperbola $xy = 6$.

a. $(2, 3)$ and $\left(\frac{3}{2}, 4\right)$

b. $(1, 5)$ and $(3, 1)$

c. $(2, 3)$ and $(3, 2)$

d. $(4, -1)$ and $(1, 6)$

3 Determine the value of k for which the line $x + y = k$ is tangent to the curve $y = \frac{4}{x}$.

a. $k = \pm 4$

b. $k = 4$ only

c. $k = \pm 2$

d. $k = 8$

Section 2: Quadratic Inequalities in One Variable

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4 Find the set of values of x satisfying the quadratic inequality $2x^2 - 5x - 3 \geq 0$.

a. $x \leq -\frac{1}{2}$ or $x \geq 3$

b. $-\frac{1}{2} \leq x \leq 3$

c. $x < -1$ or $x > 3$

d. $x \geq 3$ only

5 Solve the strict quadratic inequality $x(x + 4) < 12$.

a. $-6 < x < 2$

b. $x < -6$ or $x > 2$

c. $-2 < x < 6$

d. $0 < x < 3$

6 Find the range of values of x for which the expression $9 - x^2$ is strictly positive.

a. $-3 < x < 3$

b. $x < -3$ or $x > 3$

c. $x < 3$

d. $0 < x < 9$

Section 3: Rational Inequalities and Sign Analysis

7 Solve the rational inequality $\frac{x-2}{x+3} \leq 0$.

a. $-3 < x \leq 2$

b. $-3 \leq x \leq 2$

c. $x < -3$ or $x \geq 2$

d. $x \leq 2$

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8 Find the complete solution set for the inequality $\frac{2x+1}{x-1} > 1$.

a. $x < -2$ or $x > 1$

b. $-2 < x < 1$

c. $x > 1$ only

d. $x > -2$

9 Solve the inequality $\frac{1}{x} < 2$ for all non-zero real x.

a. $x < 0$ or $x > \frac{1}{2}$

b. $0 < x < \frac{1}{2}$

c. $x > \frac{1}{2}$ only

d. $x < \frac{1}{2}$

Section 4: Systems with Modulus and Geometric Applications

10 Find the values of x satisfying the simultaneous conditions $|x - 3| \leq 4$ and $x^2 - 4x > 0$.

a. $-1 \leq x < 0$ or $4 < x \leq 7$

b. $-1 \leq x \leq 7$

c. $0 < x < 4$

d. $4 < x \leq 7$ only

11 The perimeter of a right-angled triangle is 30 cm and its hypotenuse is 13 cm. Find the lengths of the two legs.

a. 5 cm and 12 cm

b. 6 cm and 11 cm

c. 7 cm and 10 cm

d. 8 cm and 9 cm

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12 Find the range of values of x where the line $y = 3x - 2$ lies entirely above the parabola $y = x^2 - 2x + 4$.

a. $2 < x < 3$

b. $x < 2$ or $x > 3$

c. $-3 < x < -2$

d. No such interval exists

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